

HENRY DOWD

Ireland · +353 83 032 1641 · henry@dowd.ie
linkedin.com/in/henrypdowd · github.com/hpdowd

PROFILE

Final-year BSc in Pure Mathematics & Computer Science at Maynooth University (graduating May 2026), with a year abroad at the University of Ottawa. Run a two-node k3s cluster on a single Proxmox host, built up from individual components and managed entirely through GitOps – ArgoCD reconciles the cluster against a git repository, with Sealed Secrets, Longhorn storage, MetalLB, and Traefik ingress. Comfortable with Linux administration, networking, and container orchestration, and document architectural decisions and incidents as I go. Targeting graduate roles in platform engineering, DevOps, SRE, or systems administration.

TECHNICAL SKILLS

Containers & GitOps Kubernetes (k3s), Docker, Helm, ArgoCD, Longhorn, MetalLB, Traefik, Sealed Secrets
Linux & Virtualisation Arch Linux, Proxmox VE, LXC, ZFS, systemd, shell scripting, SSH key management
Networking TCP/IP, DNS (split-horizon), DHCP, WireGuard, Cloudflare Tunnels, reverse proxies / ingress, SSL/TLS
Observability & Backup VictoriaMetrics, Grafana, Alertmanager, restic, Backblaze B2
Languages & Tools Python, Java, C, Bash, SQL; Git, Gitea, Let's Encrypt, Technitium DNS, $\LaTeX$

PROJECTS

Self-Hosted Infrastructure & Home Lab · Personal

2024 – Present

- Run a two-node k3s cluster on a single Proxmox host, managed end to end with GitOps: ArgoCD reconciles the cluster against a git repository using the app-of-apps pattern, so the repo is the single source of truth. github.com/hpdowd/homelab
- Built the platform from individual components — MetalLB for load balancing, Traefik for ingress, Longhorn for replicated storage, and Sealed Secrets to keep encrypted credentials in git.
- Expose services publicly through a Cloudflare Tunnel with no inbound ports open, and resolve the same hostnames internally with split-horizon DNS (Technitium) so LAN traffic stays in-house.
- Added a VictoriaMetrics, Grafana, and Alertmanager stack for metrics and alerting, with offsite restic backups to Backblaze B2; documented decisions as ADRs and incidents as post-mortems.

Final Year Project — Paraphrase Detection · Maynooth University

2025 – 2026

- Built a dual-branch classifier fusing BERT cosine similarity, Zhang–Shasha tree edit distance, Jaccard overlap, and length ratio; evaluated on the MSRP corpus. github.com/hpdowd/paraphrase_detector
- Implemented in Python with HuggingFace Transformers, PyTorch, and scikit-learn.
- Performed error analysis comparing structural vs. semantic methods, with confusion matrices, score distributions, and parse-tree examples of failure modes.

PDF-Sorter — Workplace Automation Tool · Python

2025

- Regex-based tool to auto-sort and rename PDFs by filename and content metadata, with OCR fallback; deployed during HR placement at Roscommon County Council. github.com/hpdowd/PDF-Sorter

EXPERIENCE

Summer Student — Human Resources · Roscommon County Council

Jun – Sep 2025

- Handled data entry, records management, and financial processing across employee files and payroll-adjacent administration.
- Identified a repetitive document-sorting bottleneck and built a custom Python tool (PDF-Sorter) to automate the workflow.

EDUCATION

BSc Computational Thinking · Maynooth University

Sep 2022 – May 2026

Pure Mathematics & Computer Science

Relevant modules: Operating Systems, Communications & Concurrency; Algorithms & Data Structures; Programming Languages & Compilers; Theory of Computation; Software Verification.

Study Abroad · University of Ottawa

Sep 2024 – May 2025

CERTIFICATIONS & ADDITIONAL

Fortinet NSE 1–3 (2026) · Full driving licence · CTYI alumnus